

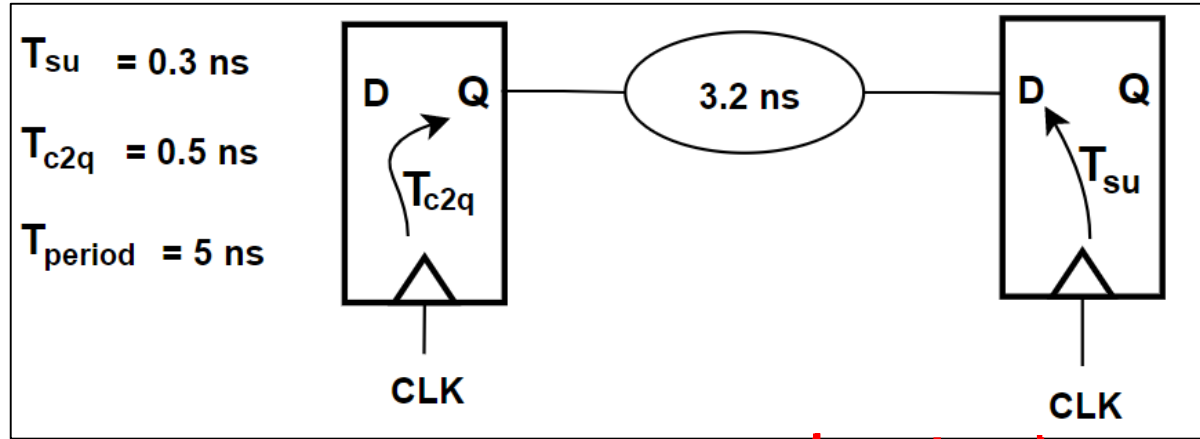
Example 1

- Assumptions

1. Interconnect Delays not considered
2. Clock Uncertainty not considered
3. One Operating Condition

- Required

1. Is this path timing is met or not? **YES MET**
2. if met calculate the positive slack and if not calculate the negative slack



$$T_{c2q} + T_{\text{comb}} + T_{su} = 4 \text{ ns} < T_{\text{period}}$$
$$T_{\text{slack}} = 5 - 4 = 1 \text{ ns}$$

+ve slack

Example 2

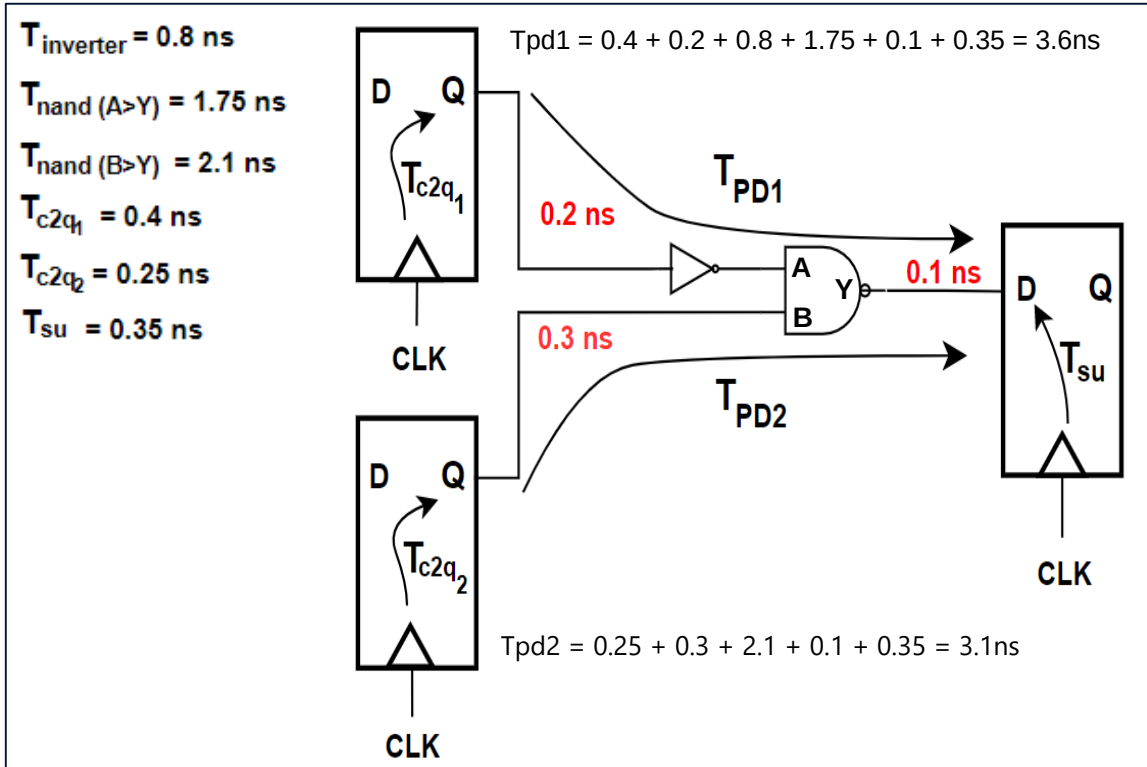
Assumptions

1. **Interconnect Delays is considered**
2. Clock Uncertainty not considered
3. One Operating Condition

Required: -

1. Calculate Tpd1
2. Calculate Tpd2
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths

$$T_{\text{period}} \geq 3.6\text{ns}$$
$$T_{\text{period_least}} = 3.6\text{ns}$$



Example 3

Assumptions

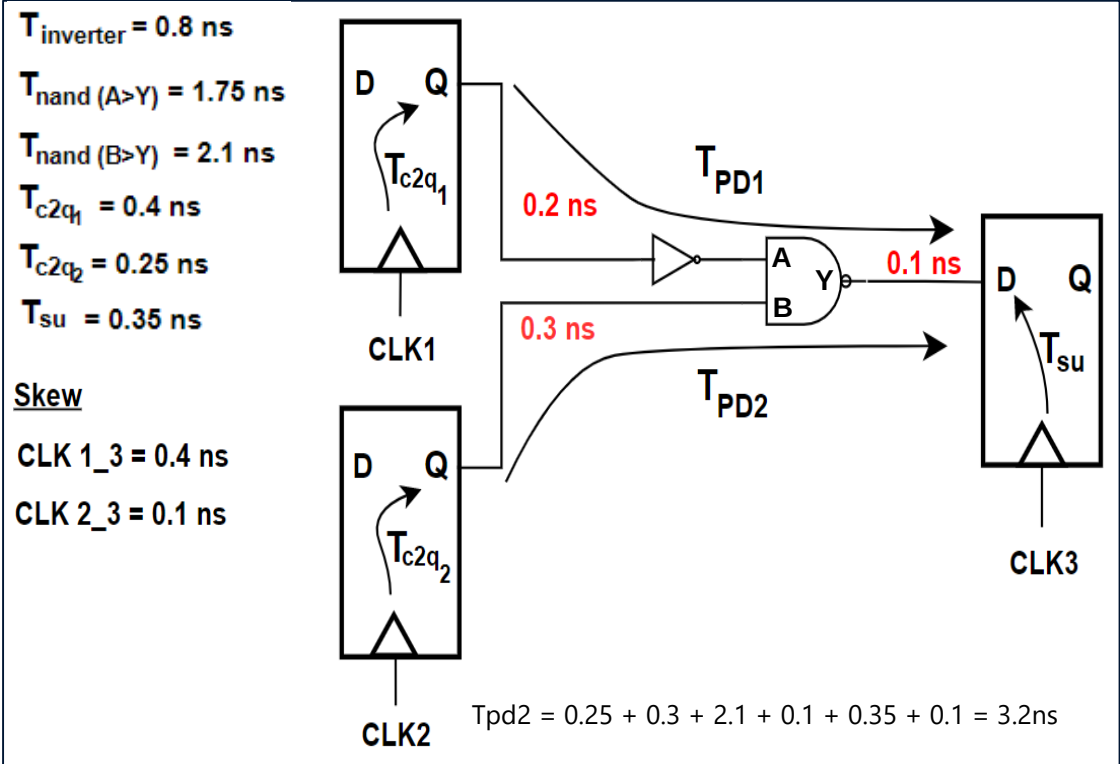
1. **Interconnect Delays is considered**
2. **Clock Uncertainty is considered**
3. **One Operating Condition**

Required: -

1. Calculate T_{pd1}
2. Calculate T_{pd2}
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths

$$T_{period_least} = 4ns$$

$$T_{pd1} = 0.4 + 0.2 + 0.8 + 1.75 + 0.1 + 0.35 + 0.4 = 4ns$$



Example 4

$$T_{pd3} = 0.4 + 0.8 + 1.7 + 1.2 = 4.1\text{ns}$$

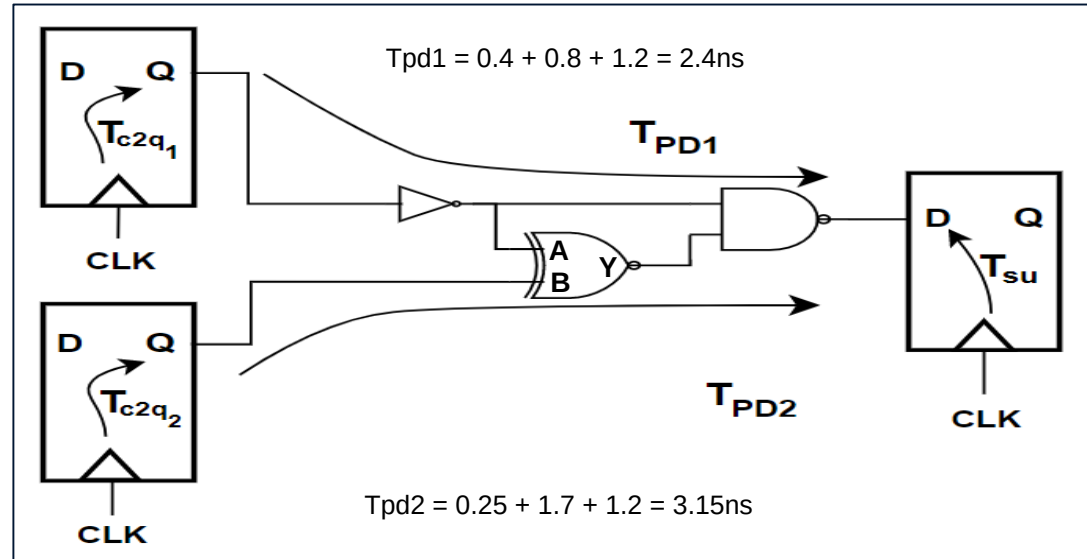
Assumptions

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

Required: -

1. Calculate T_{pd1}
2. Calculate T_{pd2}
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths

$$T_{\text{period_Least}} = 4.1 + 0.3 = 4.4\text{ns}$$



Delay	Inverter	XNOR	NAND	Tcq1	Tcq2	Tsu
Max	0.8	1.7	1.2	0.4	0.25	0.3
Min	0.65	1.55	1.1	0.3	0.2	0.2